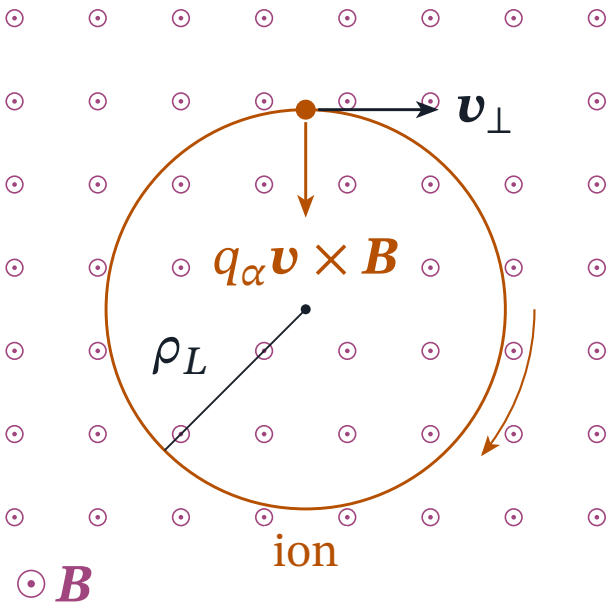
Single-particle motion

Christopher Albert

5.1 Gyration: force balance



Lorentz force $\boldsymbol{F} = q_\alpha \boldsymbol{v} \times \boldsymbol{B}$

$\boldsymbol{v} \perp \boldsymbol{B}$ $|\boldsymbol{F}| = |q_\alpha| v_\perp B$

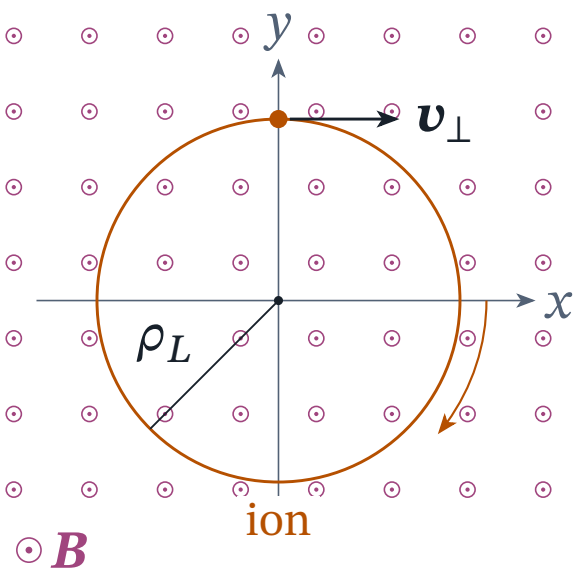
centrifugal = Lorentz $\dfrac{m_\alpha v_\perp^2}{\rho_L} = |q_\alpha| v_\perp B$

$$\omega_c = \frac{q_\alpha B}{m_\alpha}, \quad \rho_L = \frac{v_\perp}{|\omega_c|}$$

gyrofrequency, Larmor radius

q_α, m_α charge and mass of species α; $\omega_c = q_\alpha B / m_\alpha$ signed; $B = |\boldsymbol{B}|$; $v_\perp$ speed across $\boldsymbol{B}$

5.1 Equations of motion



$\boldsymbol{B} = B\boldsymbol{e}_z$ $\quad dv_x/dt = \omega_c v_y, \quad dv_y/dt = -\omega_c v_x$

$dv_z/dt = 0$

d/ dt $\quad d^2v_x/dt^2 = \omega_c\, dv_y/dt = -\omega_c^2 v_x$

solution $\quad v_x = v_\perp \cos(\omega_c t + \phi)$

$v_y = -v_\perp \sin(\omega_c t + \phi)$

integrate $\quad x = x_0 + (v_\perp/\omega_c)\sin(\omega_c t + \phi)$

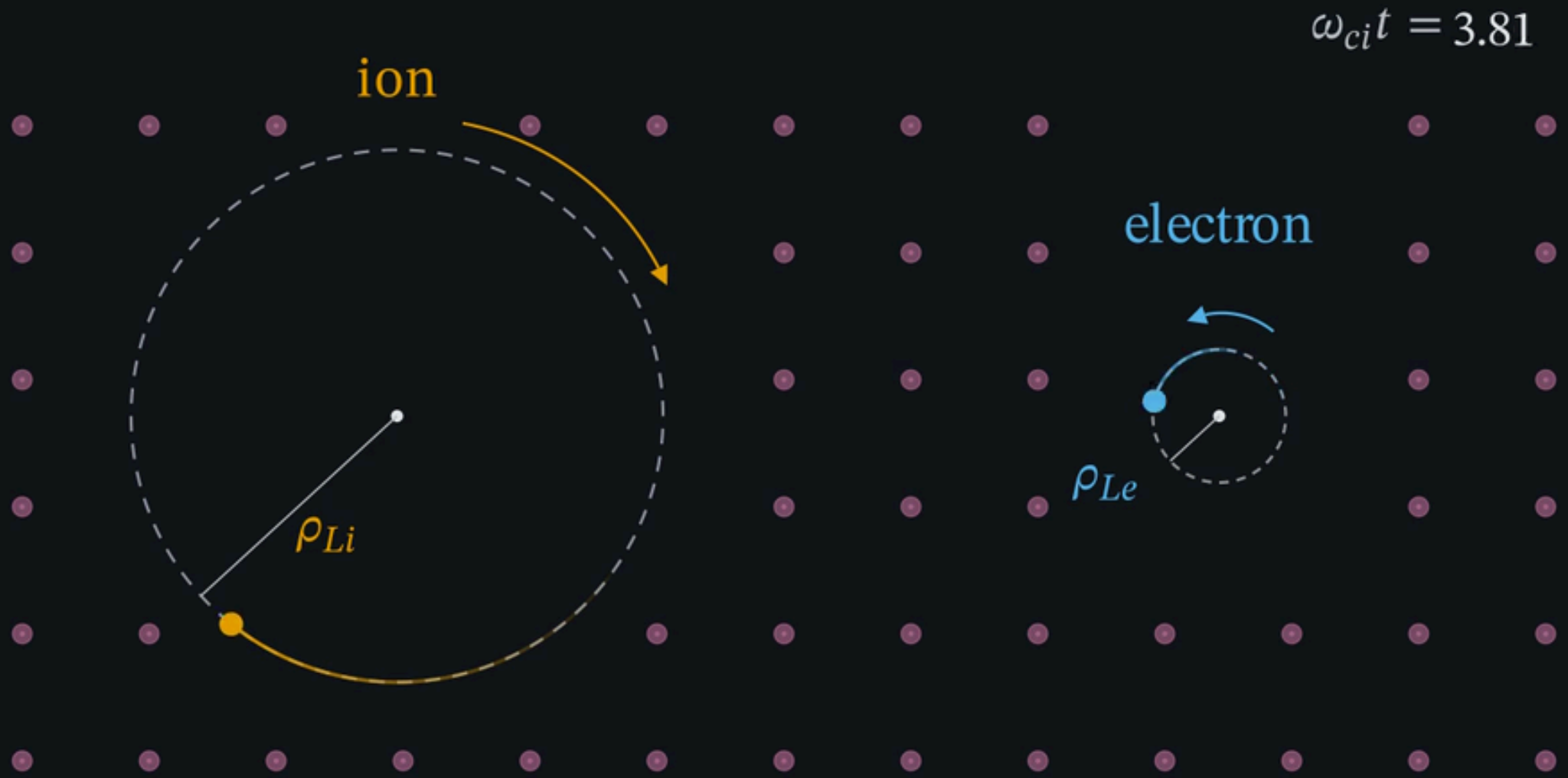
$y = y_0 + (v_\perp/\omega_c)\cos(\omega_c t + \phi)$

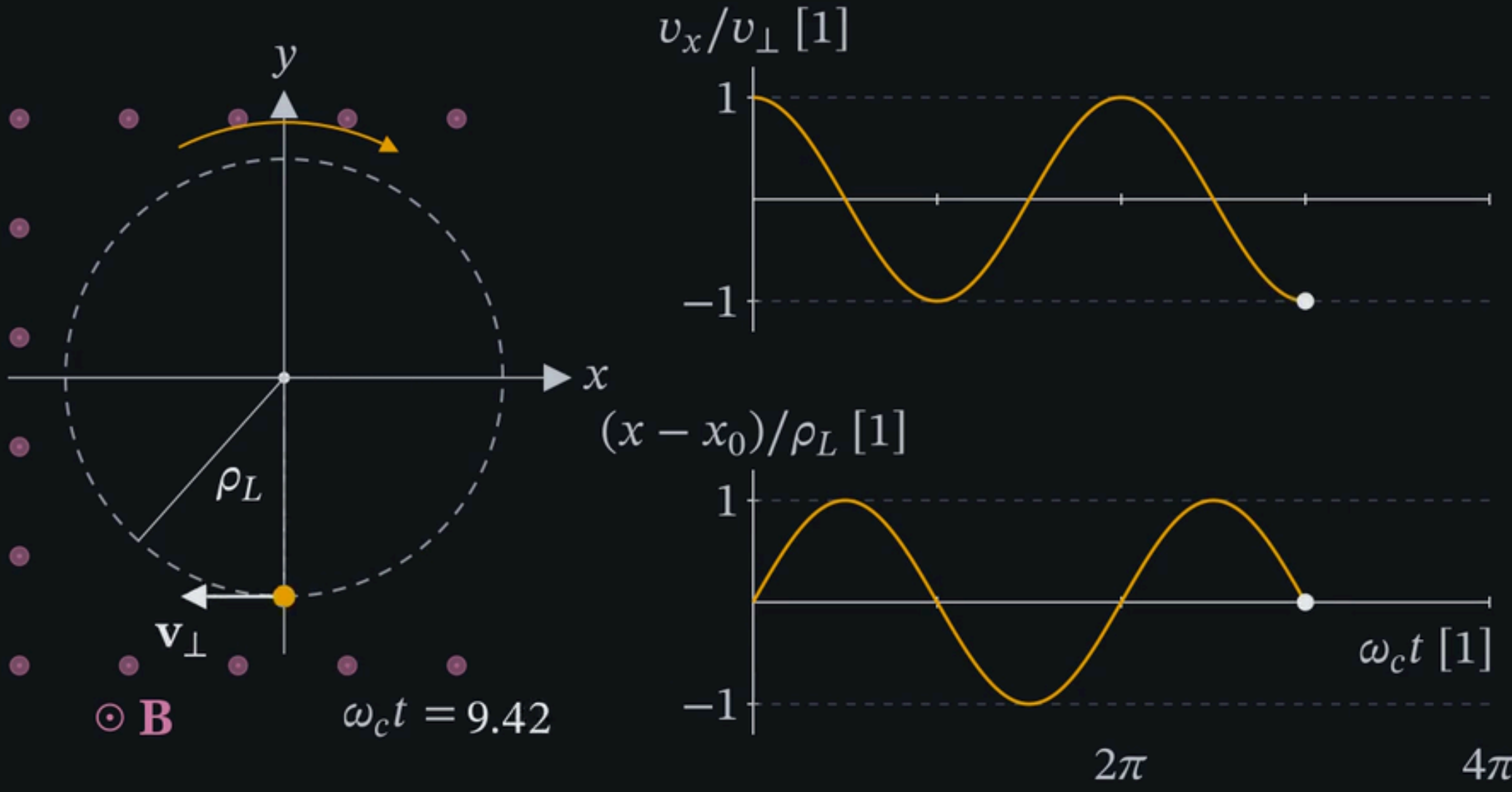
along $\boldsymbol{B}$ $\quad z = z_0 + v_\parallel t$

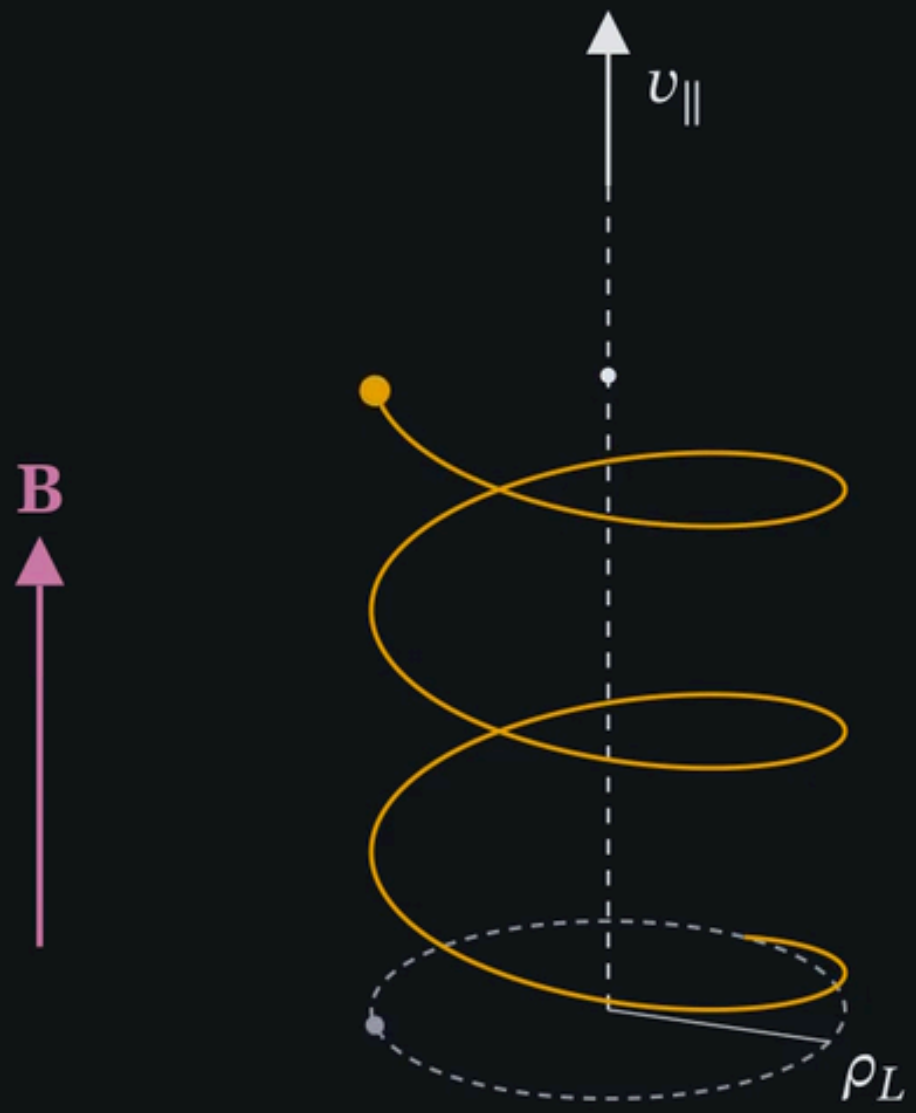
ϕ gyrophase, (x_0, y_0) guiding centre, $\rho_L = v_\perp/|\omega_c|$, $v_\parallel$ speed along $\boldsymbol{B}$

$\boldsymbol{B}$ out of the page: ions ($\omega_c > 0$) clockwise, electrons ($\omega_c < 0$) counterclockwise

F. F. Chen, *Introduction to Plasma Physics and Controlled Fusion*, 3rd ed., Springer (2016), ch. 2, doi:10.1007/978-3-319-22309-4


$\omega_{ci}t = 3.81$
ion
electron
ρ_{Li}
ρ_{Le}
$\odot$ **B**
$m_i/m_e = 16$ (H: 1836)
$T_i = T_e$


y
x
ρ_L
$\mathbf{v}_\perp$
$\odot$ **B**
$\omega_c t = 9.42$
$v_x/v_\perp$ [1]
1
−1
$(x - x_0)/\rho_L$ [1]
1
−1
$\omega_c t$ [1]
2π
4π



$\omega_c t = 16.5$

$2\pi v_\parallel / \omega_c = 1.1\,\rho_L$

5.1 Typical gyrofrequencies and Larmor radii

α	B	$\lvert\omega_c\rvert/2\pi$	ρ_L, $k_BT = 10$ eV	ρ_L, $k_BT = 10$ keV
electron	1 T	28 GHz	11 µm	337 µm
electron	5 T	140 GHz	2.1 µm	67 µm
proton	1 T	15 MHz	457 µm	14 mm
proton	5 T	76 MHz	91 µm	2.9 mm

$$\rho_L = v_{\text{th}}/|\omega_c|, \quad v_{\text{th}} = \sqrt{2k_BT/m_\alpha} \qquad T \text{ temperature, } k_B \text{ Boltzmann constant}$$

$$\rho_{Li}/\rho_{Le} = \sqrt{m_i/m_e} \approx 43 \text{ at equal } T, \quad |\omega_{ce}/\omega_{ci}| = m_i/m_e \approx 1836$$

5.1 Thermonuclear scales

hydrogen, $n_e = 10^{20}\mathrm{m}^{-3}$, $T_e = T_i = 10^8$ K ($k_B T \approx 8.6$ keV), $B = 5$ T

frequency	s^{-1}
$\lvert\omega_{ce}\rvert$	8.8×10^{11}
ω_{pe}	5.6×10^{11}
ω_{ci}	4.8×10^{8}

length	m
ρ_{Li}	2.7×10^{-3}
λ_D	6.9×10^{-5}
ρ_{Le}	6.3×10^{-5}

$$|\omega_{ce}| \sim \omega_{pe} \gg \omega_{ci}, \quad \rho_{Le} \sim \lambda_D \ll \rho_{Li} \ll \text{machine size}$$

$\omega_{pe} = \sqrt{n_e e^2/(\varepsilon_0 m_e)}$ plasma frequency, $\lambda_D = \sqrt{\varepsilon_0 k_B T_e/(n_e e^2)}$ Debye length

5.1 Magnetized plasma

$$\rho_L \ll L$$

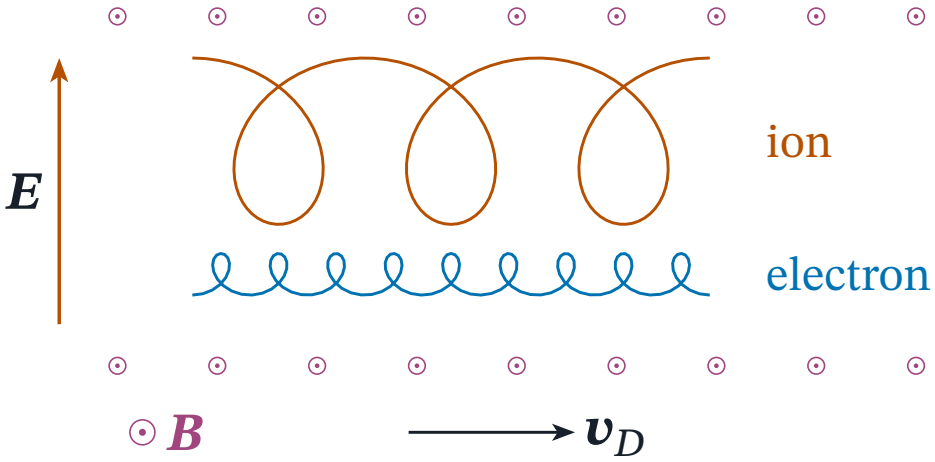
orbit small against the gradient length L

$$|\omega_c|\tau \gg 1$$

many turns between collisions, τ collision time

fusion: $\rho_{Li} \approx 3$ mm against minor radius $a \approx 1$ m; equal T: $|\omega_{ce}|\tau_e > \omega_{ci}\tau_i$

5.2 $E \times B$ drift



ansatz $\boldsymbol{v} = v_\parallel \boldsymbol{b} + \boldsymbol{v}_\perp(t) + \boldsymbol{v}_D$

gyration $\mathrm{d}\boldsymbol{v}_\perp / \mathrm{d}t = (q_\alpha/m_\alpha)\boldsymbol{v}_\perp \times \boldsymbol{B}$

remainder $0 = \boldsymbol{E}_\perp + \boldsymbol{v}_D \times \boldsymbol{B}$

$\times \boldsymbol{B}$ $0 = \boldsymbol{E} \times \boldsymbol{B} - B^2 \boldsymbol{v}_D$

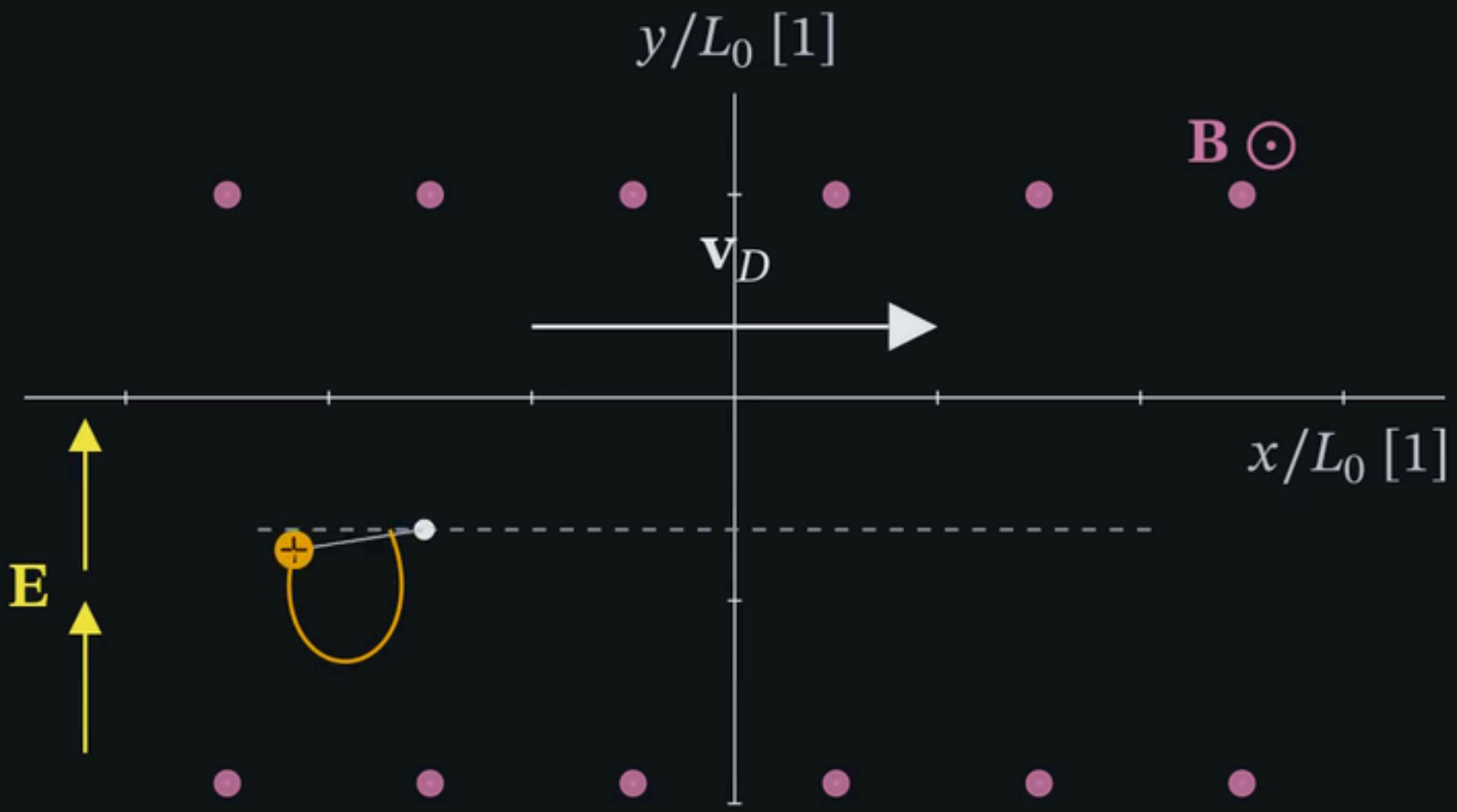
$$\boldsymbol{v}_D = \frac{\boldsymbol{E} \times \boldsymbol{B}}{B^2}$$

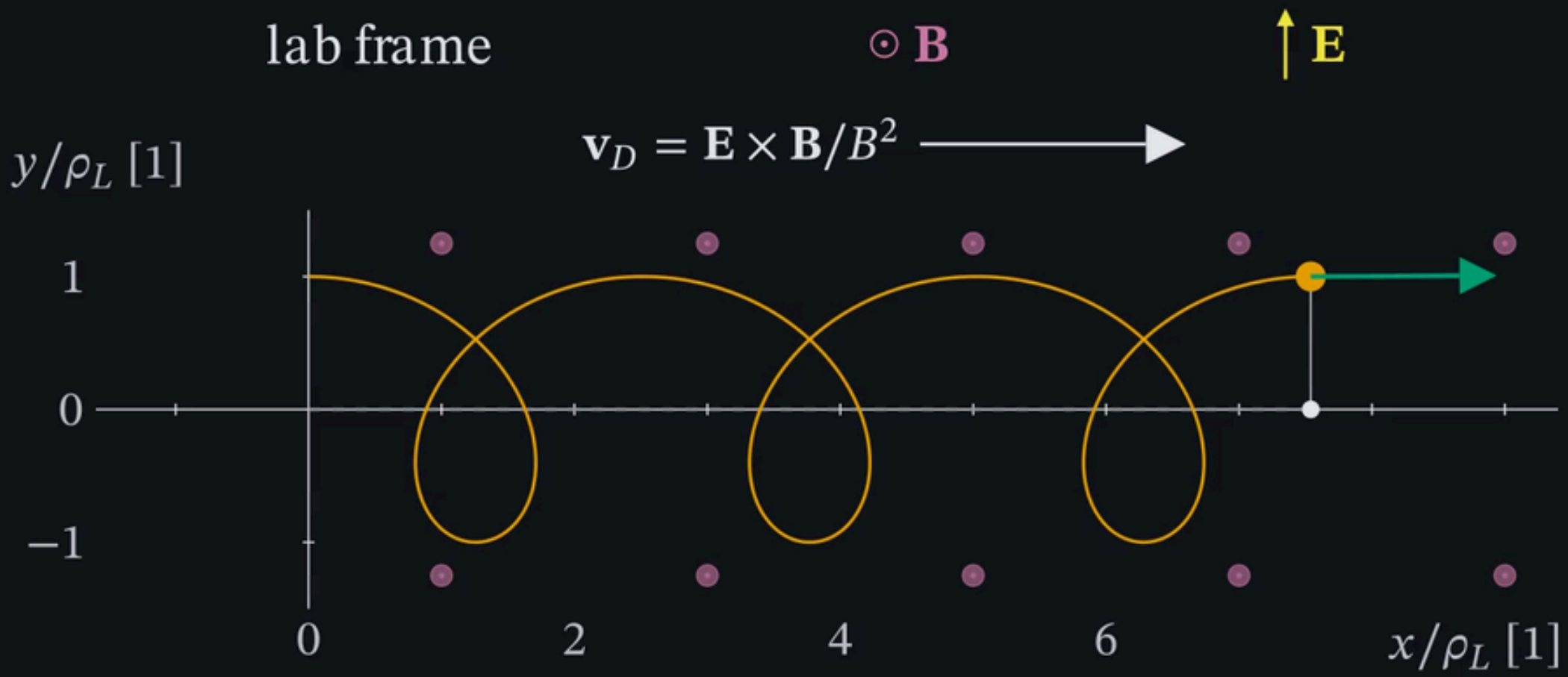
independent of q_α and m_α

drift frame $\boldsymbol{E}' = \boldsymbol{E} + \boldsymbol{v}_D \times \boldsymbol{B} = 0$ exact: gyration + uniform drift ($\boldsymbol{E} \perp \boldsymbol{B}$, $E/B \ll c$)

along $\boldsymbol{b}$ $m_\alpha \, \mathrm{d}v_\parallel / \mathrm{d}t = q_\alpha E_\parallel$ free acceleration

$\boldsymbol{b} = \boldsymbol{B}/B$, $\boldsymbol{v}_D$ drift velocity, $\boldsymbol{v}_\perp(t)$ gyration, $\boldsymbol{E}_\perp$, $E_\parallel$ field across and along $\boldsymbol{B}$


y/L_0 [1]
B ⊙
$\mathbf{v}_D$
x/L_0 [1]
E



$v_\perp - v_D \leq |\mathbf{v}| \leq v_\perp + v_D$

$\omega_c t = 18.8$

5.2 General force drift

force $\boldsymbol{F} \perp \boldsymbol{B}$ $\quad \boldsymbol{F} + q_\alpha \boldsymbol{v}_D \times \boldsymbol{B} = 0$

$\times \boldsymbol{B}$ $\quad \boldsymbol{v}_D = \dfrac{\boldsymbol{F}}{q_\alpha} \times \dfrac{\boldsymbol{B}}{B^2}$

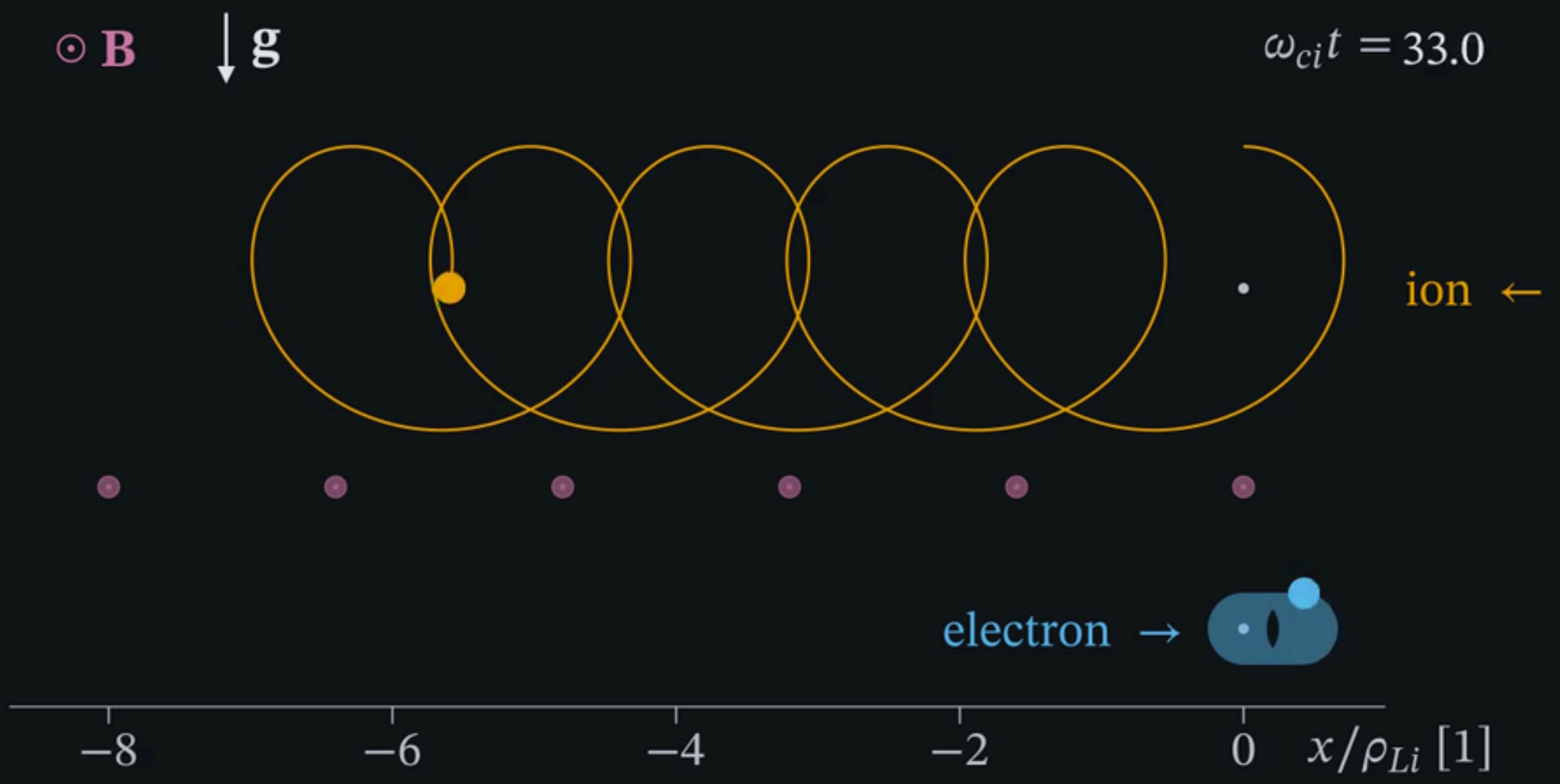
$\boldsymbol{F} = q_\alpha \boldsymbol{E}$ $\quad \boldsymbol{v}_D = (\boldsymbol{E} \times \boldsymbol{B})/B^2$ same for all species: no charge separation

$\boldsymbol{F} = m_\alpha \boldsymbol{g}$ $\quad \boldsymbol{v}_D = \dfrac{m_\alpha}{q_\alpha} \dfrac{\boldsymbol{g} \times \boldsymbol{B}}{B^2}$ ions and electrons opposite

current $\quad \boldsymbol{j} = \sum_\alpha n_\alpha q_\alpha \boldsymbol{v}_D = n(m_i + m_e) \dfrac{\boldsymbol{g} \times \boldsymbol{B}}{B^2}$

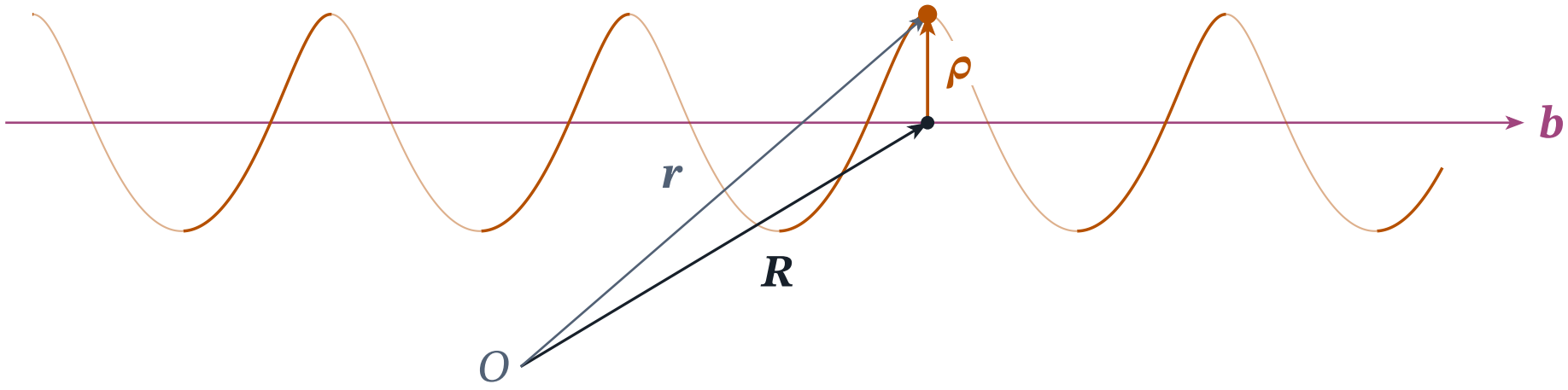
$\boldsymbol{F}$ constant force, $\boldsymbol{g}$ gravitational acceleration, n_α density of species α, hydrogen $n_i = n_e = n$

notebook: particle orbits and drifts



$$\mathbf{v}_D = (m_\alpha/q_\alpha)(\mathbf{g}\times\mathbf{B})/B^2$$

5.3 Guiding centre



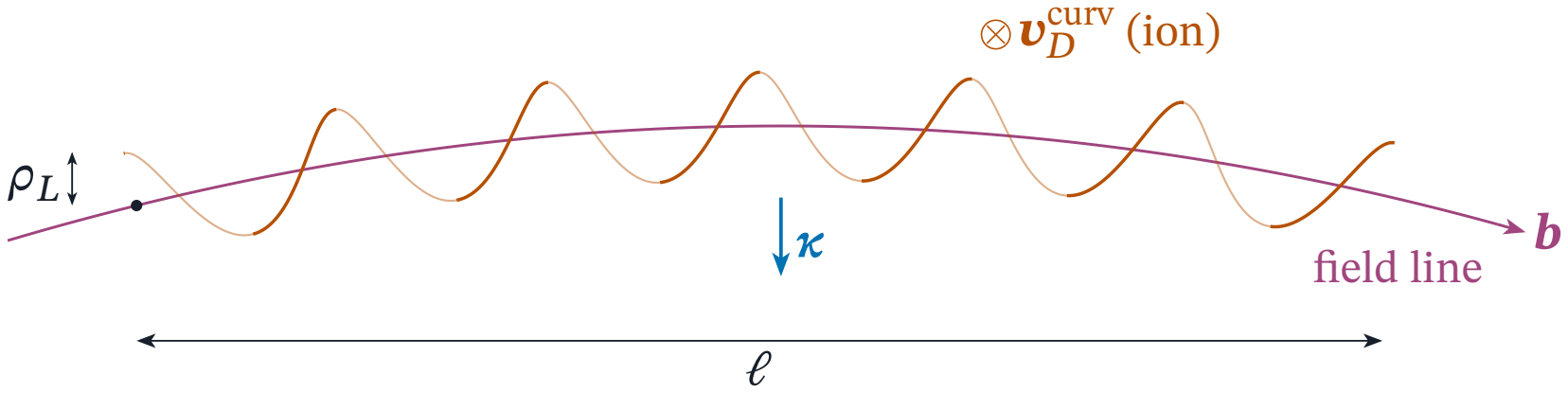
position $\boldsymbol{r} = \boldsymbol{R} + \boldsymbol{\rho}, \quad \boldsymbol{\rho} = (\boldsymbol{b} \times \boldsymbol{v})/\omega_c$

uniform $\boldsymbol{B}$ $\dot{\boldsymbol{\rho}} = \boldsymbol{v}_\perp \quad \Rightarrow \quad \dot{\boldsymbol{R}} = v_\parallel \boldsymbol{b}$

slow variation $\varepsilon = v/(|\omega_c| L) \ll 1$

$\boldsymbol{R}$ guiding centre, $\boldsymbol{\rho}$ gyration vector, L field variation length, v characteristic particle speed

5.3 Why follow magnetic field lines?



differentiate $\dot{\boldsymbol{R}} = v_\parallel \boldsymbol{b} - (\mathrm{d}(\boldsymbol{b}/\omega_c)/\,\mathrm{d}t) \times \boldsymbol{v}$

field varies over L $\mathrm{d}(\boldsymbol{b}/\omega_c)/\,\mathrm{d}t = O(v/(|\omega_c|L))$

$$\dot{\boldsymbol{R}} = v_\parallel \boldsymbol{b} + O(\varepsilon v)$$

field-line following at leading order

$\varepsilon = v/(|\omega_c|L) \ll 1$; for $v_\parallel \approx v_\perp$, this is of order ρ_L/L

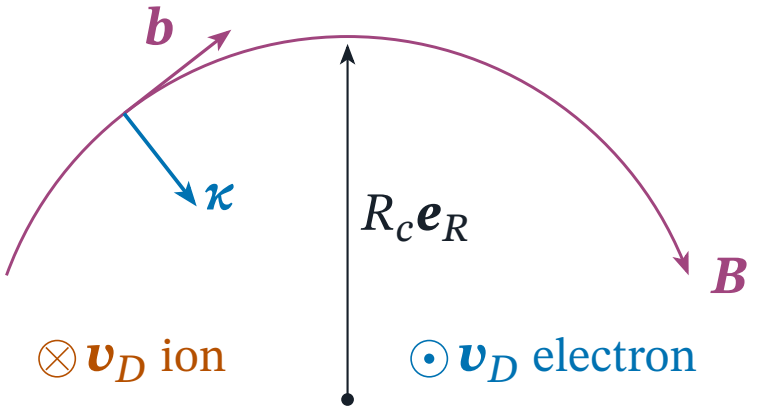
5.3 Curvature drift

centrifugal force

$$\boldsymbol{F}_c = \left(m_\alpha v_\parallel^2\right)/R_c\,\boldsymbol{e}_R$$

general force drift

$$\boldsymbol{v}_D = (\boldsymbol{F}\times\boldsymbol{B})/\left(q_\alpha B^2\right)$$

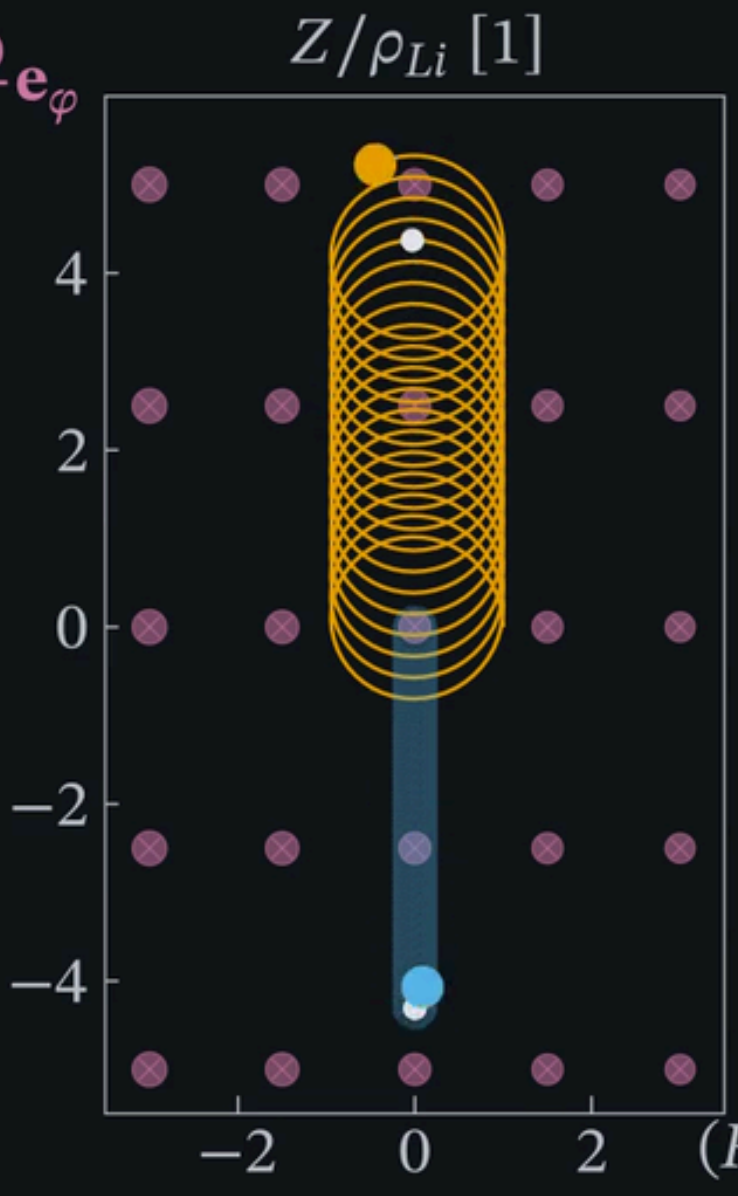


$$\boldsymbol{v}_D^{\text{curv}} = \left(m_\alpha v_\parallel^2\right)/(q_\alpha B R_c)\,\boldsymbol{e}_R\times\boldsymbol{b}$$

first-order drift

R_c field-line radius of curvature, $\boldsymbol{e}_R$ outward; $v_D^{\text{curv}}/v = O(\varepsilon)$ when $R_c \approx L$

$\otimes\, \mathbf{B} = B_0 \frac{R_0}{R} \mathbf{e}_\varphi$



$\varphi_{\text{ion}} = 2.9$ [1]

ion ↑

$$\mathbf{v}_D = \frac{m_\alpha}{q_\alpha BR}\left(v_\parallel^2 + \frac{1}{2}v_\perp^2\right)\mathbf{e}_R \times \mathbf{b}$$

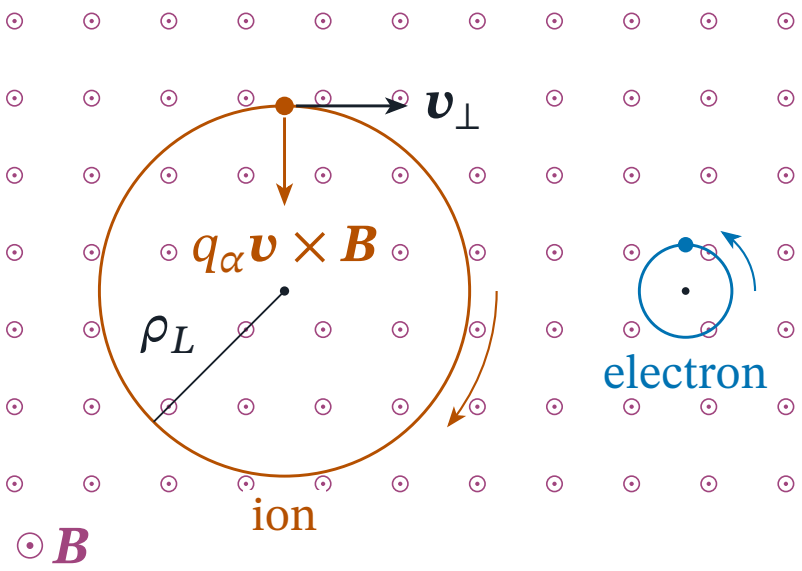
R major radius

electron ↓

5.3 Magnetic moment

orbit current $I = |q_\alpha||\omega_c|/(2\pi)$

orbit area $S = \pi\rho_L^2$



$$\mu = IS = \left(m_\alpha v_\perp^2\right)/(2B)$$

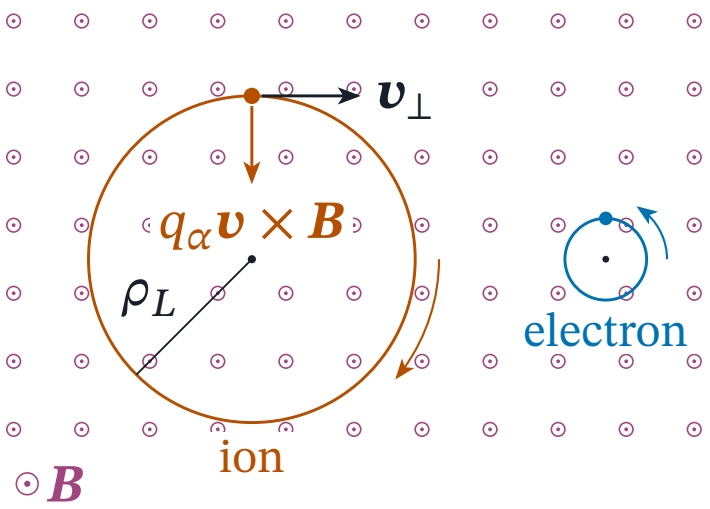
diamagnetic

Orbital dipole $\boldsymbol{\mu} = -\mu \boldsymbol{b}$: both positive and negative charges produce a moment opposite to $\boldsymbol{B}$.

5.3 Why is μ nearly constant?

fast gyration $T_c = 2\pi/|\omega_c|$

gyroaction $J = E_\perp/|\omega_c| = (m_\alpha/|q_\alpha|)\mu$



$J \approx \text{const.} \quad \Rightarrow \quad \mu \approx \text{const.}$

adiabatic invariant of a fast periodic motion

The magnetic field must vary little during one gyroperiod; the instantaneous particle moment may have small oscillations.

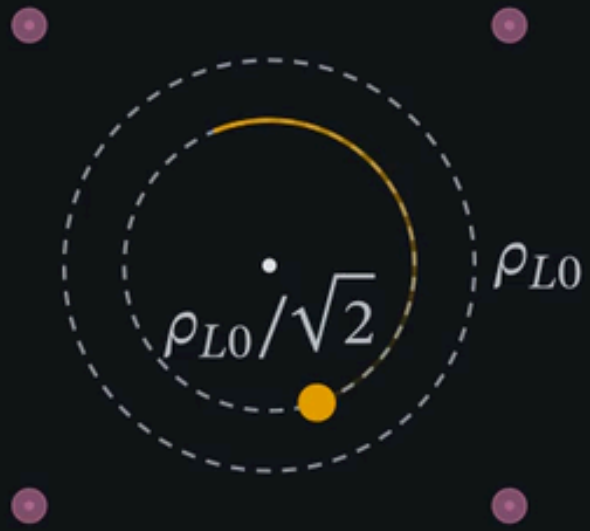
$\odot\ \mathbf{B}(t)$

$\longrightarrow\ \mathbf{E} = -\frac{1}{2}\dot{B}\,\mathbf{e}_z \times \mathbf{r}$

$B/B_0 = \frac{3}{2} + \frac{1}{2}\tanh(t/\tau_B)$

$\omega_{c0}\tau_B = 20,\ q_\alpha > 0$

$\omega_{c0} = q_\alpha B_0/m_\alpha$



$\omega_{c0}t = 160$

end: $B/B_0 = 1.999$

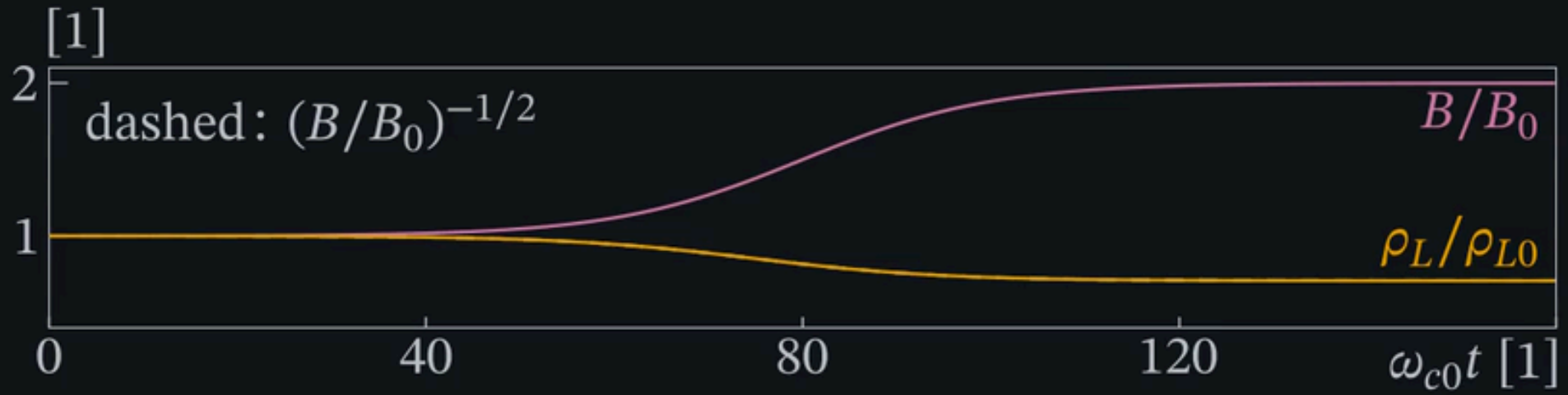
$\rho_L/\rho_{L0} = 0.7073$

$(B/B_0)^{-1/2} = 0.7073$

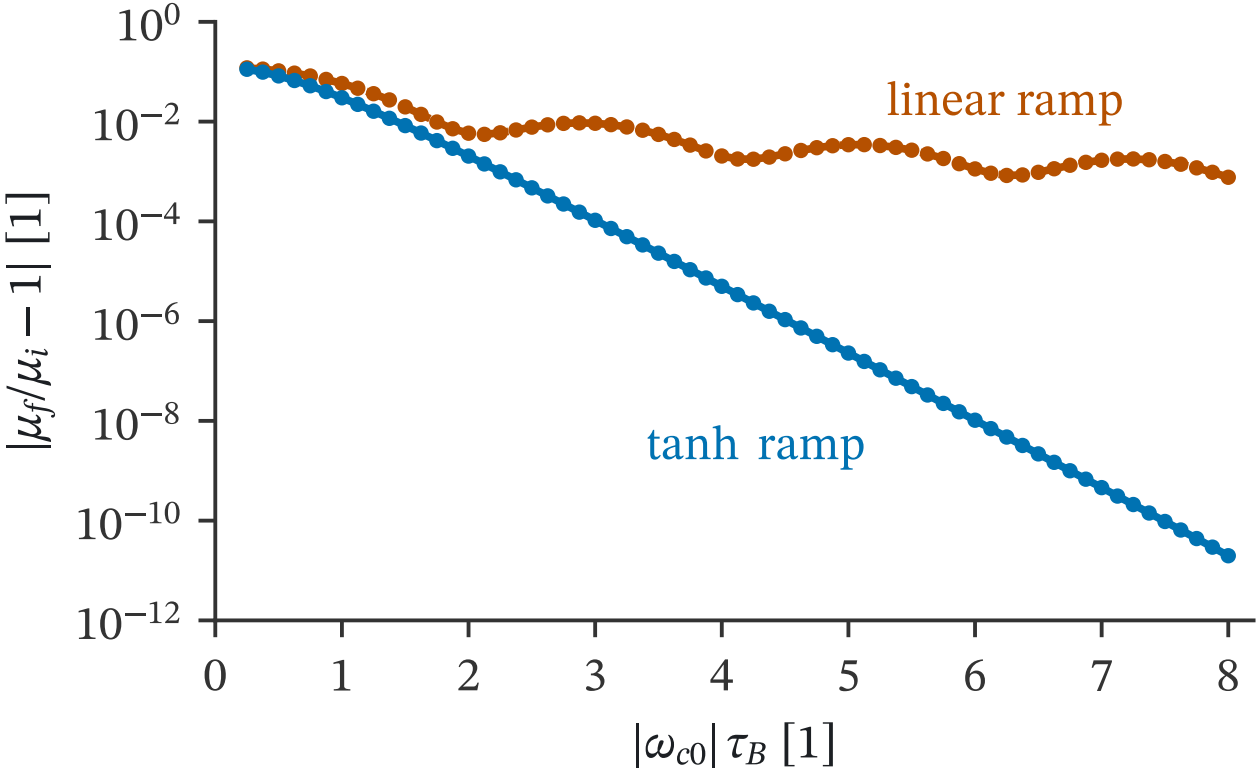
$|\mu/\mu_i - 1| \le 8 \times 10^{-5}$

$|\psi/\psi_i - 1| \le 5 \times 10^{-4}$

$\psi = \pi\rho_L^2 B$



5.3 Adiabatic invariance: numerical example



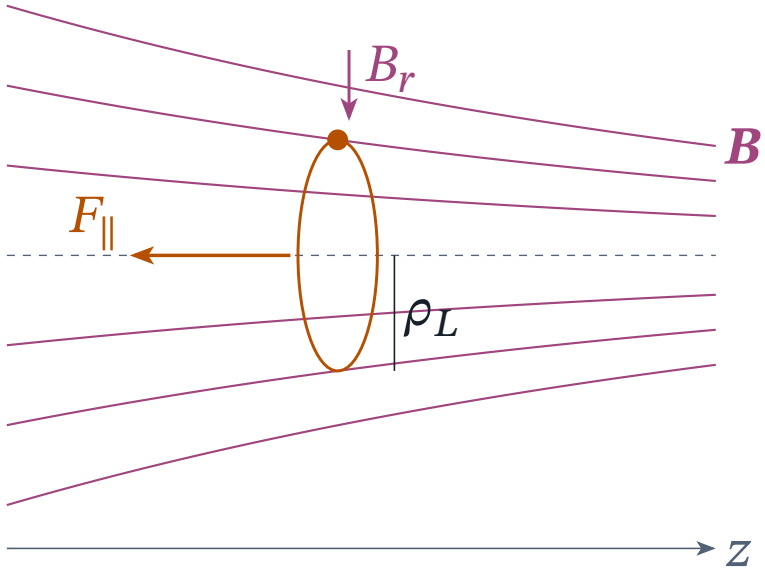
Slower field variation ⇒ better conservation of μ

Illustration: field strength doubles; μ_i and μ_f are the moments before and after the ramp.

5.3 Magnetic-moment energy and force

energy $H = \left(m_\alpha v_\parallel^2\right)/2 + \mu B$

moment force $\boldsymbol{F}_\mu = -\mu \boldsymbol{\nabla} B$



$$F_{\mu,\parallel} = -\mu \boldsymbol{b} \cdot \boldsymbol{\nabla} B, \quad \boldsymbol{F}_{\mu,\perp} = -\mu \boldsymbol{\nabla}_\perp B$$

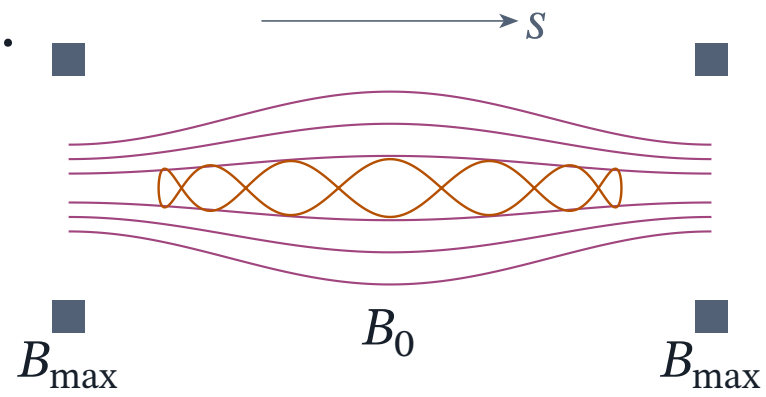
mirror force and grad-B force

μ held constant to leading order; static magnetic field, $\boldsymbol{E} = 0$

5.3 Magnetic mirror: reflection

energy $H = (m_\alpha v_\parallel^2)/2 + \mu B = \text{const.}$

turning point $v_\parallel = 0$



$$B_m = H/\mu$$

parallel energy converted into perpendicular energy

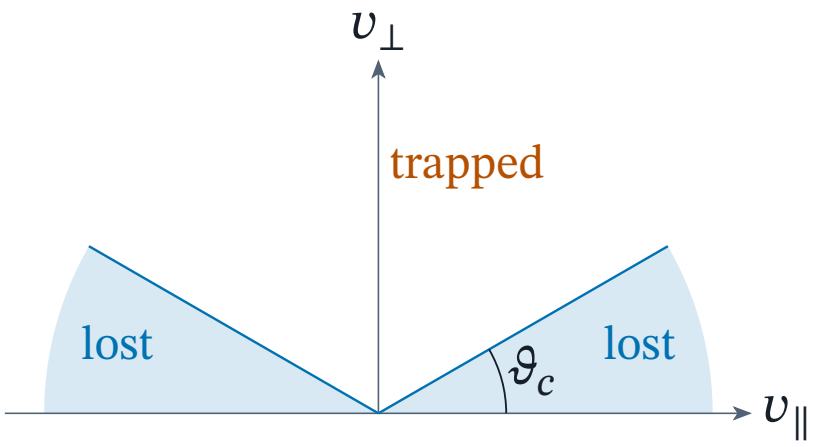
B_m magnetic field at reflection, H conserved energy, μ adiabatically invariant

5.4 Mirror ratio and loss cone

pitch angle $\sin\vartheta = v_\perp/v$

μ const., $\boldsymbol{E} = 0$ $\sin^2\vartheta/B = \text{const.}$

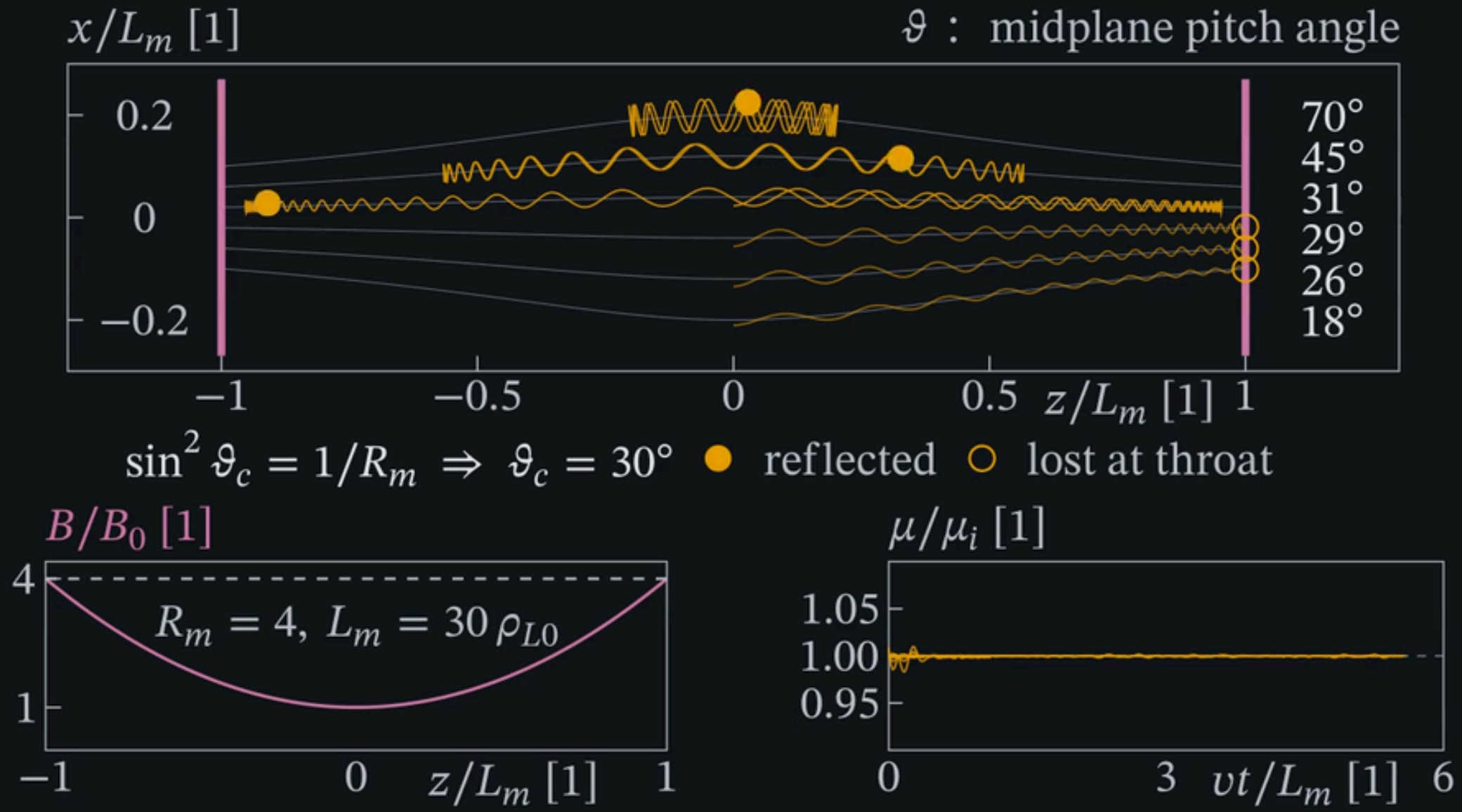
trapped $B_0/\sin^2\vartheta_0 < B_{\max}$



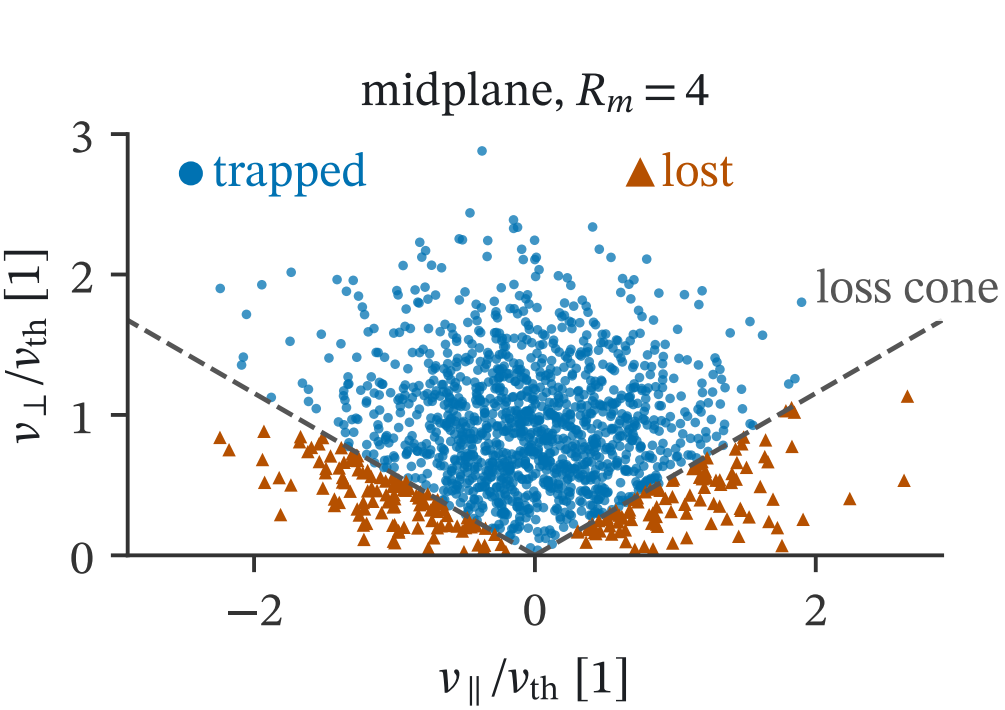
$$\sin^2\vartheta_c = 1/R_m$$

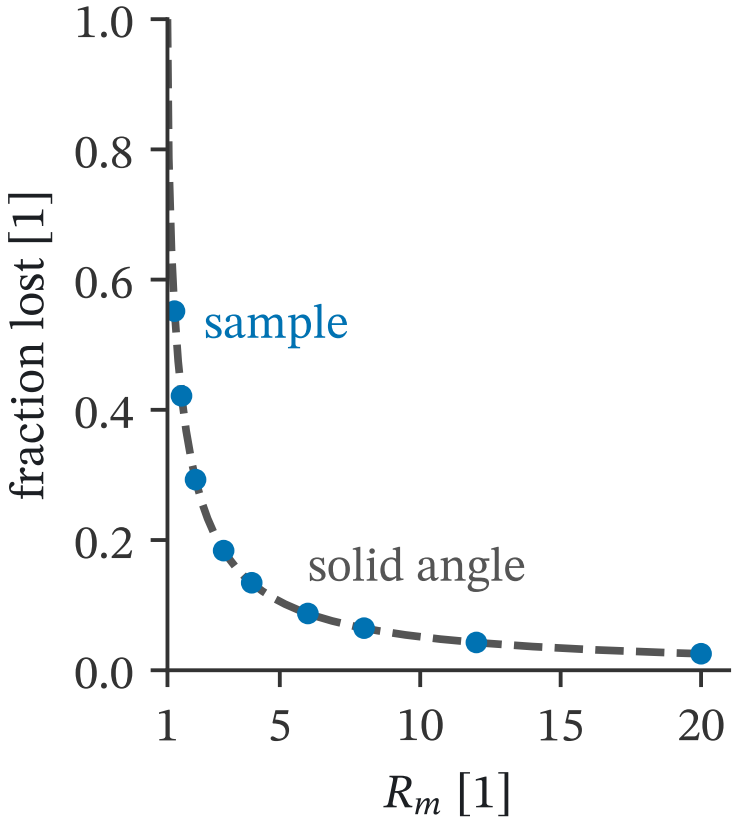
$R_m = B_{\max}/B_0$ mirror ratio

ϑ_0 pitch angle at the midplane, ϑ_c loss-cone angle; speed v constant in a static magnetic field


x/L_m [1]
ϑ : midplane pitch angle
0.2
0
−0.2
70°
45°
31°
29°
26°
18°
−1
−0.5
0
0.5
z/L_m [1]
1
$\sin^2 \vartheta_c = 1/R_m \Rightarrow \vartheta_c = 30°$
reflected
lost at throat
B/B_0 [1]
4
1
$R_m = 4,\ L_m = 30\,\rho_{L0}$
−1
0
z/L_m [1]
1
μ/μ_i [1]
1.05
1.00
0.95
0
3
vt/L_m [1]
6

5.4 Loss cone in velocity space





$$v_\parallel^2(z) = v^2 - v_{\perp 0}^2 B(z)/B_0$$

$$\text{lost} = 1 - \cos\vartheta_c = 1 - \sqrt{1 - 1/R_m}$$ isotropic velocity distribution

$v_{\perp 0}$ at midplane, $R_m = B_{\max}/B_0$, L_m mirror half-length

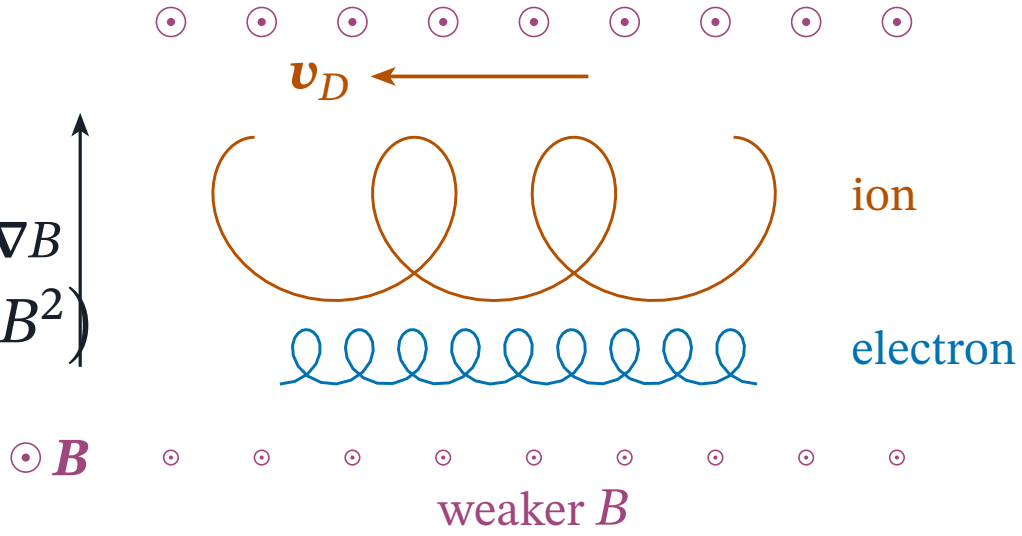
5.4 Grad-B drift

perpendicular force

$$\boldsymbol{F}_{\mu,\perp} = -\mu \boldsymbol{\nabla}_\perp B$$

general force drift

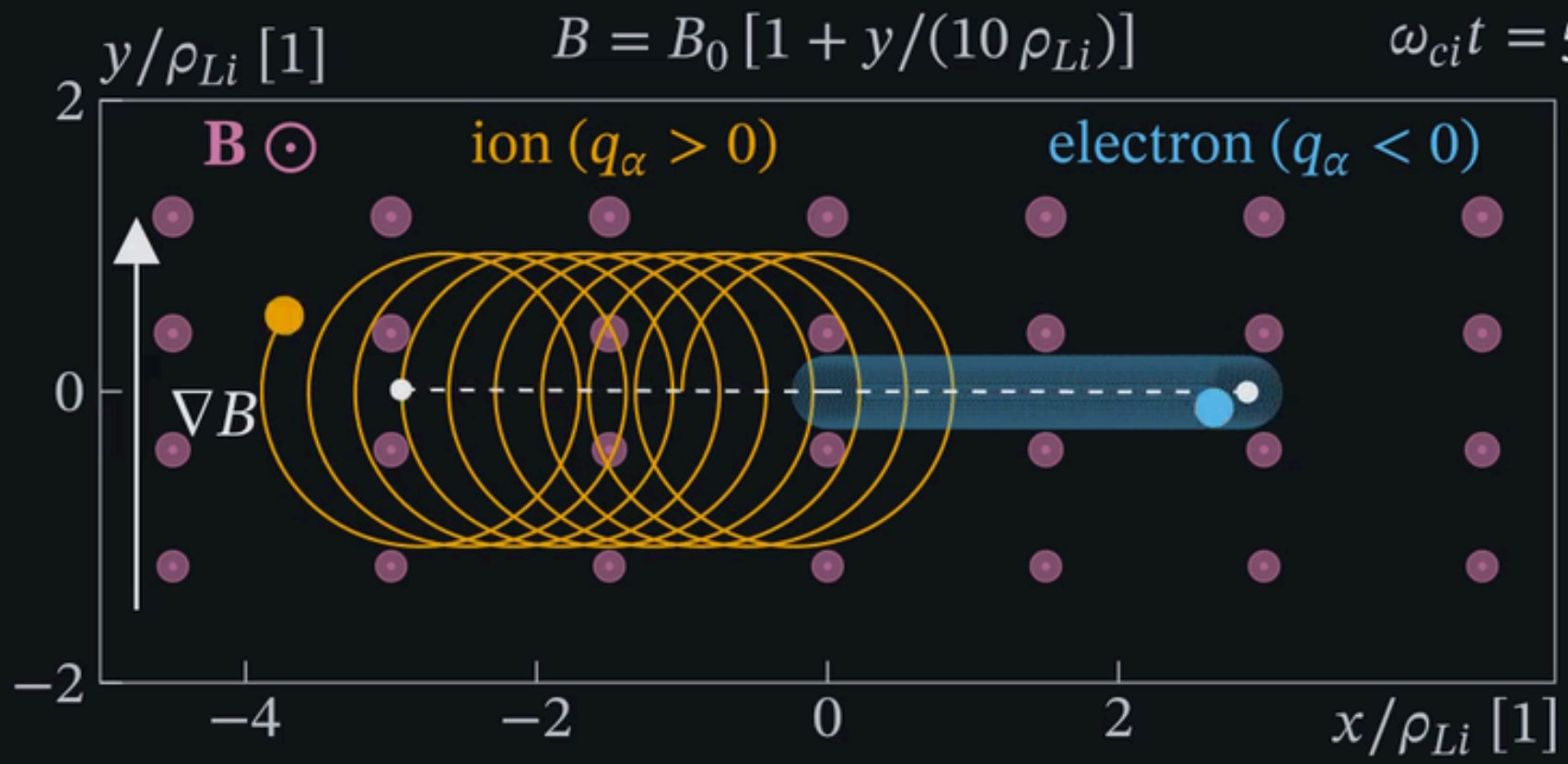
$$\boldsymbol{v}_D = (\boldsymbol{F} \times \boldsymbol{B})/(q_\alpha B^2)$$



$$\boldsymbol{v}_D^{\nabla B} = \mu/(q_\alpha B^2)\, \boldsymbol{B} \times \boldsymbol{\nabla} B$$

grad-B drift

The parallel component of the same force $-\mu \nabla B$ is the mirror force.



$v_{D,x}^{\nabla B}/(\rho_{Li}\omega_{ci})$ [1] orbit | $-\mu\,\partial_y B/(q_\alpha B)$ (dev.)

ion −0.0506 | −0.0500 (+1.3 %)

electron +0.0501 | +0.0500 (+0.1 %)

$m_i/m_e = 16,\ T_i = T_e$

5.4 Guiding-centre motion: results

$$\text{static } \boldsymbol{B},\ \boldsymbol{E} = 0;\ \varepsilon = \frac{v}{|\omega_c| L} \ll 1$$

field-line motion
$$\dot{\boldsymbol{R}} = v_\parallel \boldsymbol{b} + O(\varepsilon v)$$

magnetic moment
$$\mu = \frac{m_\alpha v_\perp^2}{2B} \approx \text{const.}$$

energy
$$H = \frac{m_\alpha v_\parallel^2}{2} + \mu B \approx \text{const.}$$

mirror force
$$m_\alpha \dot{v}_\parallel \approx -\mu \boldsymbol{b} \cdot \boldsymbol{\nabla} B$$

curvature drift
$$\boldsymbol{v}_D^{\text{curv}} = \frac{m_\alpha v_\parallel^2}{q_\alpha B R_c} \boldsymbol{e}_R \times \boldsymbol{b}$$

grad-B drift
$$\boldsymbol{v}_D^{\nabla B} = \frac{\mu}{q_\alpha B^2} \boldsymbol{B} \times \boldsymbol{\nabla} B$$

μ invariant to leading order after gyroaveraging; mirror acceleration and perpendicular drifts are first-order effects.

5.5 Polarization drift

slow $\boldsymbol{E}_\perp(t)$ $$\boldsymbol{v} = v_\parallel \boldsymbol{b} + \frac{\boldsymbol{E}\times\boldsymbol{B}}{B^2} + \boldsymbol{v}_D^{\text{pol}} + \boldsymbol{v}_\perp$$

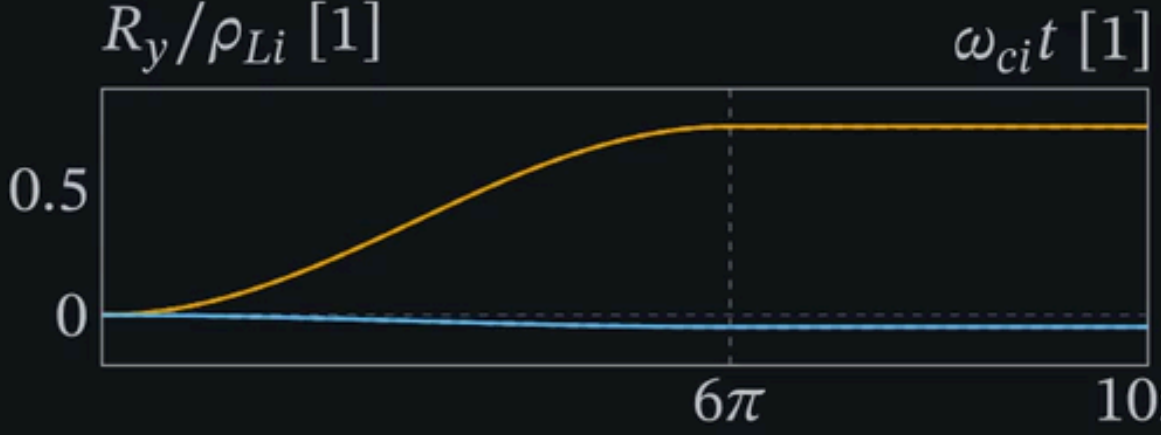
drift $$\boldsymbol{v}_D^{\text{pol}} = \frac{m_\alpha}{q_\alpha B^2}\frac{\partial \boldsymbol{E}_\perp}{\partial t}$$

current $$\boldsymbol{j}_{\text{pol}} = \sum_\alpha n_\alpha q_\alpha \boldsymbol{v}_D^{\text{pol}} = \frac{\sum_\alpha n_\alpha m_\alpha}{B^2}\frac{\partial \boldsymbol{E}_\perp}{\partial t}$$

validity $$\omega \ll |\omega_c|$$

$\boldsymbol{E}_\perp$ field across $\boldsymbol{B}$, ω its rate of change, n_α density of species α; carried mainly by ions





$\omega_{ci}t = 31.4$

$\odot$ **B**

E(t) $E_1/B = 0.75\, v_{\perp i}$

$m_i/m_e = 16,\ T_i = T_e$

$$\mathbf{v}_D^{\mathrm{pol}} = \frac{m_\alpha}{q_\alpha B^2}\frac{d\mathbf{E}_\perp}{dt}$$

$\Delta R_y/\rho_{Li}$ [1]: orbit | $\frac{m_\alpha E_1}{q_\alpha B^2}$

ion +0.7500 | +0.7500

electron −0.0469 | −0.0469

dashed: $\frac{m_\alpha}{q_\alpha B^2}E(t)$

$x_D = \int_0^t E/B\, dt$

Drifts

drift	$\boldsymbol{v}_D$	ions vs electrons	mass
$E \times B$	$(\boldsymbol{E} \times \boldsymbol{B})/B^2$	same	none
gravity	$(m_\alpha/q_\alpha)(\boldsymbol{g} \times \boldsymbol{B})/B^2$	opposite	$\propto m_\alpha$
grad-B	$(\mu/q_\alpha)(\boldsymbol{B} \times \boldsymbol{\nabla} B)/B^2$	opposite	$\propto m_\alpha v_\perp^2$
curvature	$m_\alpha v_\parallel^2 (\boldsymbol{e}_R \times \boldsymbol{b})/(q_\alpha B R_c)$	opposite	$\propto m_\alpha v_\parallel^2$
polarization	$\left(m_\alpha/\left(q_\alpha B^2\right)\right) \partial \boldsymbol{E}_\perp/\partial t$	opposite	$\propto m_\alpha$

$\boldsymbol{v}_D = (\boldsymbol{F}/q_\alpha) \times \boldsymbol{B}/B^2$ for a force $\boldsymbol{F}$; opposite drifts carry currents

5.6 Cyclotron resonance

rotating $\boldsymbol{E}$ $\quad E_{\mathrm{cw}} = E_x + iE_y = E_0 e^{-is\omega t}$ $\quad s = +1$ clockwise, $s = -1$ counterclockwise

$v_{\mathrm{cw}} = v_x + iv_y$ $\quad \mathrm{d}v_{\mathrm{cw}}/\,\mathrm{d}t = -i\omega_c v_{\mathrm{cw}} + (q_\alpha/m_\alpha)E_{\mathrm{cw}}$

from rest $\quad v_{\mathrm{cw}} = \dfrac{q_\alpha E_0}{m_\alpha} e^{-i\omega_c t} \dfrac{e^{i\Delta t} - 1}{i\Delta}, \quad \Delta = \omega_c - s\omega$

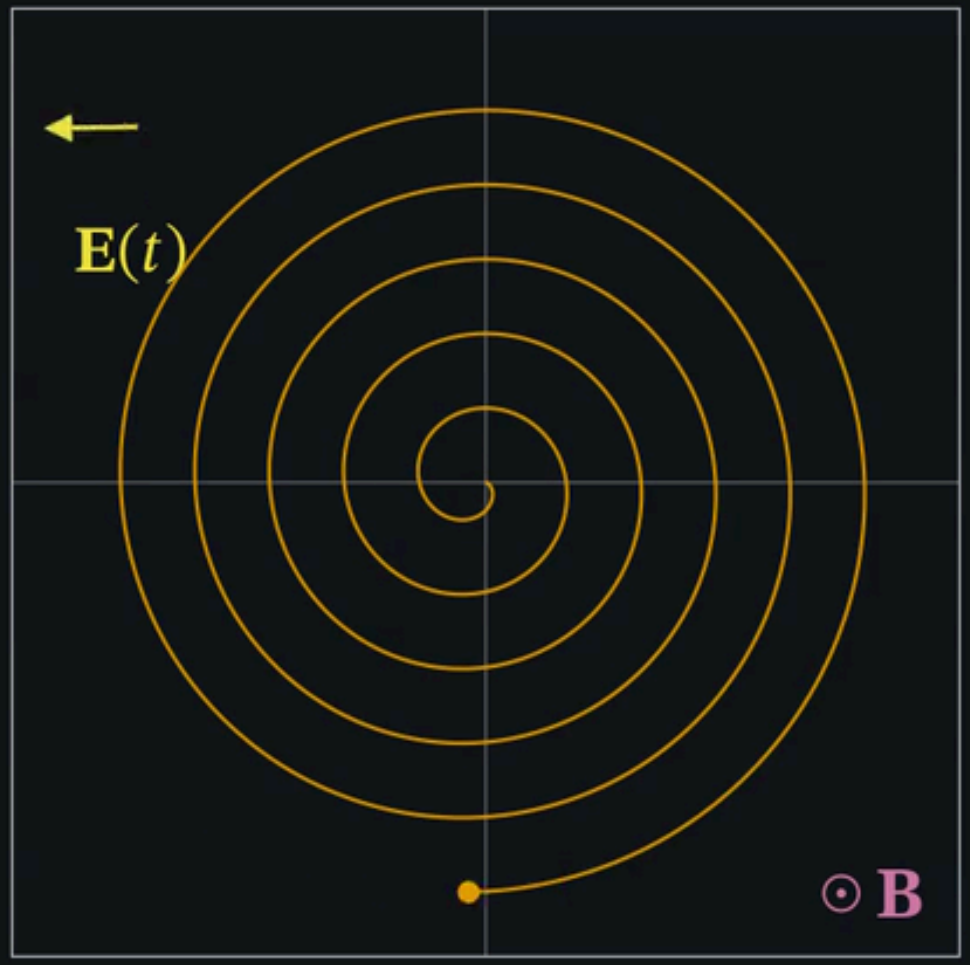
modulus $\quad v_\perp = |v_{\mathrm{cw}}| = \dfrac{|q_\alpha| E_0}{m_\alpha} \left| \dfrac{2\sin(\Delta t/2)}{\Delta} \right|$

$$\Delta = 0: \quad v_\perp = \frac{|q_\alpha| E_0}{m_\alpha} t, \quad W = \frac{q_\alpha^2 E_0^2}{2m_\alpha} t^2$$

resonance $s\omega = \omega_c$: ions clockwise, electrons counterclockwise

$\boldsymbol{B} = B\boldsymbol{e}_z$ out of the page, $\omega_c = q_\alpha B/m_\alpha$ signed, Δ detuning: off resonance $v_\perp$ beats with period $2\pi/|\Delta|$; W kinetic energy

ion, **E** clockwise, $\omega = \omega_{ci}$



$(x, y)\ |\omega_c|B/E_0\ [1]$

$|\omega_c|t = 34.5$

